Design and Implementation of 915MHz RF Energy Harvesting Sensor Node

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***Abstract* — This work presents the design and implementation of a remote environmental sensing node powered exclusively through radio-frequency (RF) energy harvesting. A 915 MHz Software-Defined-Radio (SDR) module transmits an RF power signal captured by a high-gain rectenna into a low-loss capacitor until sufficient energy for system operation is stored. When a 1mJ threshold is reached, the harvesting circuit powers a low-energy microcontroller (MCU) to measure ambient temperature and humidity. The node then transmits the raw data over Bluetooth low energy to a handheld device for decoding and display.**

***Index Terms* — Bluetooth low energy, energy harvesting, energy management, radio frequency, remote sensing, wireless power transmission.**

I. Introduction & Motivation

Environmental monitoring systems are an essential part of research, industry, and housing, delivering important data about temperature, humidity, air quality, and other aspects of the environment. Traditional sensors rely heavily on batteries or active components which limit flexibility of deployment, increase maintenance costs, and dramatically reduce their long-term viability. In remote or hard to access environments, replacing or recharging batteries can be impractical, making conventional power solutions more challenging for large-scale or long-term sensing applications. These obstacles motivate the exploration of alternative power sources that can enable autonomous, maintenance-free operation.

Radio Frequency (RF) energy harvesting has emerged as a promising solution to these limitations. Ambient and intentionally transmitted RF signals can be captured and converted into usable energy, enabling electronic systems to operate without direct connection to a power source. Recent advances in rectifier efficiency, methods of rectification, impedance matching networks, and low power microcontrollers (MCUs) have made it feasible to design sensor nodes that function solely on harvested RF energy. By utilizing RF energy harvesting architecture, sensing devices can achieve indefinite operational lifetimes, significantly reduced environmental impact, and greater implementation capabilities. This approach aligns with the growing demand for sustainable, self-powered electronics in the Internet of Things (IoT) ecosystem.

This project aims to serve as an example of a passive RF powered sensing node capable of operating solely from intentionally radiated RF energy. This begins with the development of a handheld transceiver that produces a directed beam of RF energy and can capture wirelessly transmitted data. Next, a reliable circuit capable of harvesting, storing, and converting RF energy into a stable DC supply must be constructed. Once sufficient energy has been collected, the goal is to operate an ultra-high-efficiency MCU that will power on, sample ambient environmental data, wirelessly transmit the measurement, and power off. The handheld transceiver will then capture the transmitted data, decode it, and display it.

II. System Design

A. Description of Requirements

The overall system must be able to generate an RF power signal from a handheld device, capture and store this signal as usable energy in a remote device, take a remote environmental measurement, and transmit this measurement back to the handheld device. This is most directly implemented through a combination of three systems. The first system is the handheld transceiver. The handheld transceiver must be able to generate and transmit an RF power signal, capture wirelessly transmitted sensor data, then decode and display this data. The second system is the energy harvester. The energy harvester must be able to capture an RF power signal, rectify it into usable DC power, store the harvested energy, determine when there is sufficient energy for sensor board operation, and provide a regulated DC power source. The final system is the environmental sensor. The environmental sensor must be able to power on, capture environmental sensor readings, and transmit them wirelessly as efficiently as possible. A high-level flowchart of this system is depicted in Fig. 1.

B. Handheld Transceiver

The handheld transceiver must transmit RF power, receive wireless data, and display it. The transceiver is a handheld device. It will therefore require onboard batteries

for power. A power regulator will then convert the battery voltage to a usable DC supply to power the board. A software-defined radio (SDR) will be used to generate the RF power signal. This signal will then be amplified and transmitted using a highly directional antenna to minimize power loss. To receive wireless sensor data, a second antenna and receiver must be included at a different frequency than the power transmission frequency to minimize noise. A display must be included to provide a digital readout of captured sensor data. To control the SDR, control the display, and decode the raw sensor data, an MCU must be included as well. Finally, to allow user input, a button control system must be included. The system is summarized in Fig. 2.

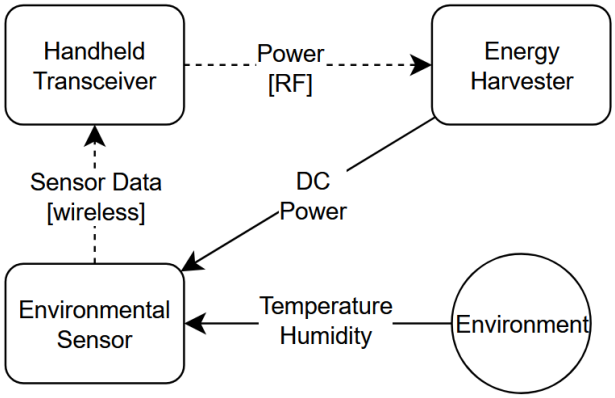


Fig. 1. High-level overview of system operations

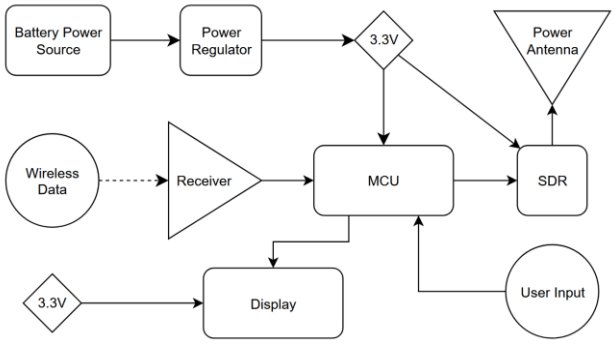


Fig. 2. Overview of transmitter board components, power flow, and data flow.

C. Energy Harvester

The purpose of the energy harvester is to capture RF energy, store it until there is sufficient energy to operate the sensor, and provide power to the sensor once this threshold is reached. The harvester will therefore require a receiving antenna and high-frequency rectifier to convert the captured RF into usable DC. This low-voltage DC signal will require an ultra-low-power harvesting IC and a storage element to convert it to usable energy. The system must then determine the energy available, and once a threshold is reached, convert it to a usable power source. The harvester will therefore require a power regulator and a method of determining available energy. The system is summarized in Fig. 3.

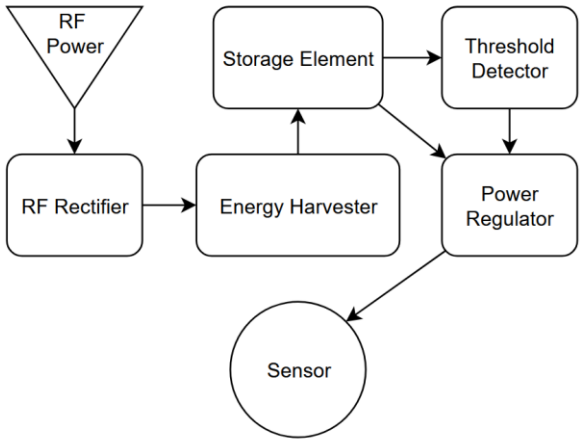


Fig. 3. Overview of energy harvesting board components and power flow

D. Environmental Sensor

The low-power sensor system must power on, capture environmental data, and wirelessly transmit it as efficiently as possible. The system will therefore require a low power environmental sensor and an ultra-low-power MCU to read from this sensor. To transmit the read data, an ultra-efficient wireless transmitter and antenna will be required. This system is summarized in Fig. 4.

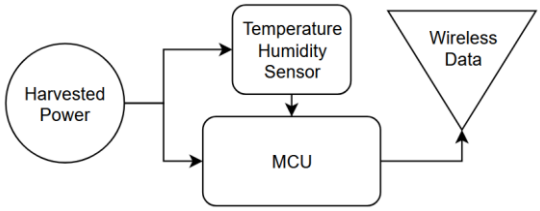


Fig. 4. Overview of low-power sensor components, power, and data flow

E. System Standards

For power transmission, a delicate balance between Free Space Path Loss (FSPL), or power attenuation over distance, and antenna size must be maintained. Lower frequency transmission allows for lower FSPL, and therefore longer transmission distances, at the cost of a larger antenna. 915MHz was selected for reasonable antenna sizes (λ=33cm), a moderate FSPL of -31dB at one meter, and crucially its location within an ISM band of the

FCC frequency spectrum. FCC ISM bands do not have access control and allow for significant power radiation. Using a Frequency Hopping Spread Spectrum (FHSS) radiation pattern of ≥ 50 channels allows for +36dBm (4W) of Effective Isotropic Radiated Power (EIRP) to be transmitted, and +30dBm (1W) at <50 channels.

For wireless communications, Bluetooth Low Energy (BLE) was selected. BLE is a short-range wireless communication protocol designed specifically for ultra-low-power operation while maintaining reliable connectivity between embedded devices. Operating in the 2.4GHz ISM band, BLE utilizes FHSS across 40 channels to minimize interference and maintain stable communication in environments with heavy RF congestion. BLE is widely used in wearables, medical sensors, and smart home devices because it enables small battery powered nodes to operate for months. For this application, BLE offers a lightweight and energy efficient method for transmitting sensor data between devices.

III. Hardware Design

A. Environmental Sensor

The most energy intensive aspects of the environmental sensor are MCU operation and BLE data transmission. The STM32WB05 MCU was chosen because of its best-in-class operational efficiency and native BLE integration, which removes the need for a discrete transmitter and associated losses. A low 1.8V minimum operating voltage further reduces V^2 loss compared to 3.3V MCUs.

For their microwatt-scale average power consumption, the SHT41 digital humidity sensor and the STS4L digital temperature sensor were selected to sample environmental data. An I^2C interface allows for simple integration and fast read time, minimizing overall energy consumption.

B. Energy Harvester

To maximize energy absorption, a highly directional narrow-band patch antenna was selected for power reception. To minimize cost, a readily available +8dBi antenna was chosen.

Energy harvesting integrated circuits generally have minimum startup voltage and minimum startup power requirements that result in effective startup impedances of approximately 1-30kΩ. For maximum power transfer, it is desirable to match this impedance to the 50Ω impedance of the harvesting antenna. This, however, is not directly possible, as harvesting ICs require DC power. Furthermore, the impedance of rectification circuits at high frequencies is dominated by diode ground shunt capacitance which limits their impedance to <500Ω. Given the large impedance mismatch of the two systems, a rectifier with as high a voltage gain as possible is desired. For this reason, a voltage-doubling full-wave rectifier topology was selected. It is possible to design rectifiers with larger voltage gains, but these require more diodes and thus incur greater diode $I*V_f$ losses.

To minimize power signal attenuation and maximize the range of operational frequencies, diodes with as small a capacitance as possible are desired. To minimize diode $I*V_f$ losses and maximize the output voltage, as small a forward voltage as possible is desired. The JDH2S02SL-L3F series of diodes were chosen for their ultra-low 0.25pF capacitance and 0.2V forward voltage drop.

To maximize possible charging distance, an energy harvesting IC with as low a startup voltage as possible is desired. With a turn-on voltage of 20mV, the LTC3108 is an attractive option; however, it has a conversion efficiency of only 50%, requires external transformers, and requires significant available power to start up. The AEM30330 was selected for its 275mV turn on voltage, 90% conversion efficiency, and 3uW startup power. Furthermore, it contains an integrated buck-boost converter that is activated when an internal threshold detector determines that there is sufficient voltage across the storage element and an integrated maximum power point tracking (MPPT) system that dynamically adjusts the apparent load of its power input pin to match the source impedance [2]. This IC contains three significant system components on one chip, greatly reducing circuit complexity and cost.

For the energy storage element, capacitors were chosen. This is due to the direct energy-voltage relation of capacitors, which allows a direct measurement of stored energy when combined with the threshold detector of the AEM30330. Kemet C4A polypropylene film capacitors were selected for their ultra-low sub-0.1% loss tangent, low memory, and ultra-high 30GΩ/μF dielectric resistance.

C. Handheld Transceiver

To provide the 915MHz power signal, the Waveshare Core1262 LoRa SDR module integrates Semtech's SX1262 transmitter IC alongside necessary 3.3V regulation, an SPI interface, and an antenna output matched to 50Ω. This allows for simple integration of the RF power transmitter that would otherwise be a complex design aspect.

For system power, standard AA batteries were selected. Being cheap and readily available, AA batteries provide a well-supported platform for portable power. These were combined with the LM2576 3.3V power regulator, chosen for simplicity of integration. The LM2576 3.3V utilizes a low switching frequency of 52kHz which improves the reliability of the regulator and decreases design complexity.

To control the transceiver, the STM32WB55 MCU was chosen. As with the environmental sensor system, native BLE support reduces complexity and allows for a cheaper product. An impressive -120dBm BLE sensitivity allows the environmental sensor system to use low power for transmission while minimizing packet loss. Furthermore, the STM32WB55 utilizes the same development environment as the STM32WB05 MCU. This drastically speeds up development time, as there is no need to learn and integrate multiple IDEs.

To display transmitted sensor data and provide a clear user interface, the Elegoo 0.96" I^2C-controlled OLED was selected. I^2C provides a simple method of interfacing with the display, and the color depth of OLED technology provides clear outputs.

For user input, a simple array of five 6mm tactile push buttons were arranged in a plus-shape. This arrangement of buttons provides a versatile input scheme for the user to control the state of the handheld transceiver.

IV. Software Design

A. Environmental Sensor

The software architecture for the environmental sensing module is optimized for low-power and time constrained operation. Because the system relies on RF energy harvesting, the software must operate efficiently with little overhead, requiring software design that is focused on short and efficient execution. The architecture is divided into sensing and wireless communication levels, each designed to operate independently.

Sensor readings will be delivered by the inter-integrated circuit (I2C) channel which comes in the form of raw data from the sensors. To make the most efficient use of the limited available energy from the harvester, environmental data will remain as raw data and will be embedded into packets for BLE transmission. Because the board runs on limited harvested energy, the firmware uses a compact format to keep radio use as short as possible. Once the energy harvester has stored over 1mJ at a voltage of 2.5V, the MCU will power on and perform its initialization. Startup takes a fixed amount of time with little room for optimization; therefore, software must be optimized elsewhere. As such, an I2C read is commanded to the SHT41 and STS4L using the minimum allowable sample time and BLE advertising starts immediately.

The BLE data is transmitted using the Eddystone-TLM format, which is a non-connectable beacon protocol commonly used for advertising data such as temperature, battery life, and other device health parameters [1]. Eddystone-TLM was chosen for its non-connectable configuration and easily formatted advertisement of data. BLE communication operates primarily in a broadcast only mode, periodically advertising temperature and humidity readings without maintaining a persistent connection to conserve energy. Because the energy harvesting circuit only powers the environmental sensor for a short period of time, the firmware must utilize short and rapid beacon transmissions that embed raw sensor readings directly into the advertising payload. The handheld transceiver scans for these advertisements and extracts the raw sensor readings without requiring the sensing module to remain connected or continuously active.

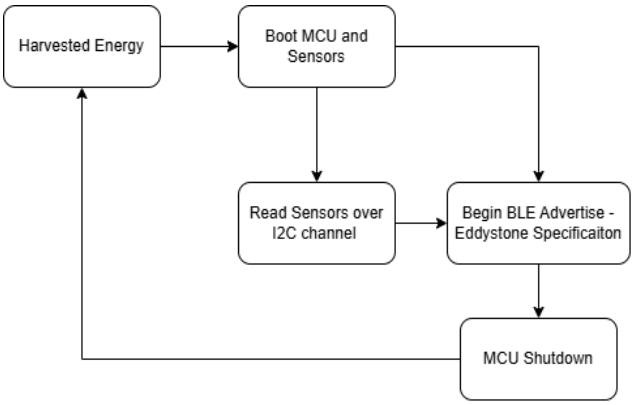


Fig. 5. Overview of Environmental Sensor Software

B. Handheld Transceiver

The handheld transceiver must transmit an RF signal using the SX1262 and display temperature and humidity data to the user. The handheld transceiver provides multiple user inputs using buttons which allow the selection of RF transmission and BLE scanning states. At the software level, the STM32WB55 must communicate with and control the SX1262 and OLED display peripherals. Semtech supplies the needed driver library for their SX1262 module, and additional firmware was developed to realize the needed FHSS scheme to transmit power within FCC limitations. Through user input, the program transitions into the power transmission state, which is managed by an interrupt that configures the SX1262 PA for +22dBm and FHSS using 60 channels. The SX1262 continues to transmit until the environmental sensor is charged, allowing it to wake, sense, and transmit measured sensor data.

While transmitting the RF power signal, the firmware will enable a Bluetooth Low Energy (BLE) scanning state, based on user input, to receive environmental sensor data. This button activated mechanism ensures that the user does

not waste transceiver battery life. The user input for BLE scanning causes the microcontroller to transition the device from its idle state into BLE scanning mode. During scanning, the handheld transceiver listens for the BLE packet sent by the environmental sensor and displays the received data through the Elegoo 0.96" OLED display.

The STM32WB55 on the handheld transceiver is responsible for processing the raw-byte sensor readings to reduce consumption of the limited energy available to the environmental sensor. The MCU extracts the telemetry frame from each received advertisement, isolates the temperature and humidity fields, then converts the raw byte values into real units. This includes applying scaling factors defined by the Eddystone specification. In the case for humidity sensor readings, the sensing board embeds custom sensor values into unused sections of the Eddystone-TLM frame. The handheld transceiver reads these specific byte offsets within the advertisement contents and converts them to represent temperature and humidity. The processed values are then forwarded to the user interface to display the environmental sensor readings.

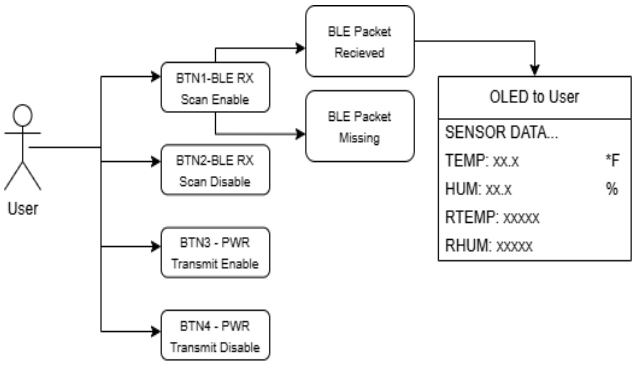


Fig. 6. Overview of Handheld Transceiver Software Design

V. System Testing

A. Energy Harvester

The energy harvester must demonstrate 6 main tasks. The antenna must (1) convert incident power to an RF signal on the PCB. This 50Ω power signal must (2) be impedance matched to the input of the rectifier circuit. The rectifier then must (3) convert the RF power signal into a DC supply for the harvester chip. The harvester chip must then (4) store this power in the capacitor, (5) detect when the voltage/energy threshold is reached, and (6) provide a stable power supply for the environmental sensor.

(1) To test the conversion of incident power into a usable 915MHz RF signal, a 2GHz bandwidth Rohde & Schwarz RTO1024 Oscilloscope was used to probe a test point included on the 915MHz transmission line. Measurement confirmed RF power absorption.

(2) To test the impedance matching of the input port, a Keysight P5004B Vector Network Analyzer (VNA) was used. The nonlinear nature of diode rectifiers necessitates measurements at multiple input powers; therefore, the S11 parameters at -10dBm, -5dBm, and 0dBm were taken. Matching circuits were then developed and tested. The best matching circuit tested was a simple short along the transmission path. This resulted in a transmission coefficient of approximately 95%.

(3) To test the functionality of the rectifier circuit, RF power was provided by a Keysight P5004B VNA. A dedicated RF source was unavailable to the authors. An SMA T-splitter was connected between the VNA output and the RF IN port of the energy harvesting circuit. The third leg of the T splitter and the output of the rectifier were connected to an RTO1024 Oscilloscope on 1MΩ termination for input and output voltage probing. Measurements confirmed the functionality of the rectifier circuit to -30dBm at 50Ω.

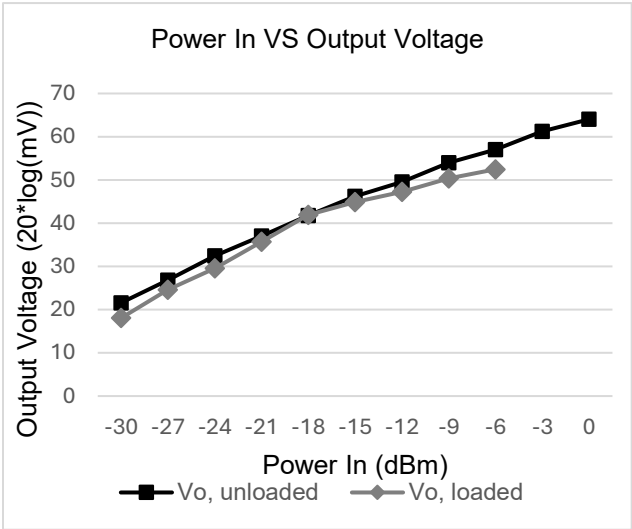


Fig. 7. Power vs rectifier output voltage. Loaded voltage could not be measured past -3 dBm due to AEM30330 charging

(4) To test energy storage in the capacitors, a 20MΩ multimeter was connected across the capacitors and a Keysight P5004B VNA was used to provide power. Conversion of DC power to stored energy was confirmed on the multimeter to -10dBm of RF input power from the VNA. Below this value, the DC output of the rectifier was below the turn-on voltage of the AEM30330 and it

therefore could not capture energy. Above this value, the sole observed difference was capacitor charge time.

(5) To test the energy threshold detection of the AEM30330, the capacitors were charged as in step (4) with the sensor board as a load attached to the energy harvester. An RTO1024 oscilloscope was connected to the capacitors, and the stored energy was tracked. Activation of the internal buck-boost converter was determined by measuring the power-out port of the energy harvester on channel 2 of the oscilloscope. Testing confirmed threshold detection at the correct voltage and simultaneous power-up of the power-out port on the energy harvester.

(6) To ensure stable output of the energy harvester, it was assembled as in step (5) and the environmental sensor was attached as a load. An RTO1024 oscilloscope was attached to the power-out port of the energy harvester and configured in one-shot mode with the trigger level set to 2.5V, the desired output voltage of the harvester circuit. Upon threshold detection, a 2.5V signal was observed on the oscilloscope for 152ms, confirming operation.

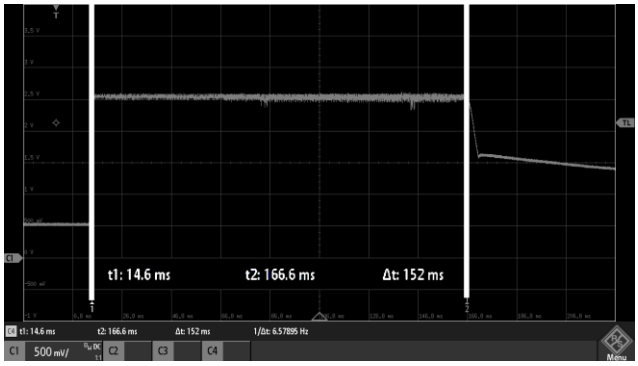


Fig. 8. Oscilloscope readout of power-out pin of the AEM30330. Horizontal bars at 14.6 ms and 166.6 ms show a Δ of 152 ms

Upon completion of all 6 steps, the energy harvester was labeled complete with full functionality and no further operational testing was performed.

B. Environmental Sensor

The environmental sensor must demonstrate 3 main tasks. It must (1) turn on under a 2.5V power signal. It must then (2) utilize the embedded MCU to poll the onboard sensors for environmental data. Finally, it must (3) enter a continuous Bluetooth beacon state broadcasting this sensor data until it is fully depleted of energy.

(1) To test activation of the environmental sensor, the MCU was configured to toggle a GPIO pin at high speed for reading by an oscilloscope. This testing confirms start-up of the MCU, the low-speed crystal oscillator, and the high-speed crystal oscillator. The environmental sensor was set up as in V: *Energy Harvester* section (6). system testing section and RTO1024 was connected to its power-out pin in one-shot mode. Channel 2 of the oscilloscope was connected to the GPIO, and testing confirmed proper output signal frequency. An LED was not used for MCU testing because of the large power draw and inability of the engineers to discern a 150ms flash. Furthermore, a single LED blink does not allow for testing of the crystal oscillators.

(2) To test proper Bluetooth transmission, the ST BLE Toolbox app was used. The environmental sensor was configured to output the raw reported sensor value of MCU temperature and display it in the app. Initial testing was performed by connecting the environmental sensor to a constant 2.5V DC supply and scanning for the Bluetooth signal with the ST BLE Toolbox app. Once transmission and data acquisition was confirmed, the environmental sensor was connected as in step (1), and the ST BLE Toolbox app was used to scan for transmission. Testing confirmed BLE transmission under the limited energy condition.

(3) To test functionality of the I^2C temperature and humidity sensors, the environmental sensor was configured to output raw reported sensor values of temperature and humidity for the ST BLE Toolbox app. Raw values are used to avoid the cost of conversion on the energy constrained sensor board. Testing was then performed as in step (2), initially on a DC supply and finally under the limited energy condition. Values of temperature and humidity were then hand-decoded and verified in multiple environments. Testing confirmed proper operation of the I^2C sensors under the limited energy condition.

Upon completion of software development and all 3 testing steps, the environmental sensor was labeled complete with full functionality, and no further testing was performed.

C. Handheld Transceiver

The handheld transceiver must demonstrate 6 main tasks. It must (1) convert power from the onboard batteries into a usable 3.3V. This power will then (2) activate the MCU and all supporting components. The MCU must then (3) program the OLED to display a user interface. Next, the MCU must (4) program the SX1262 SDR to output its maximum power and (5) radiate it from the antenna. After this, it must (6) enter a Bluetooth scanning mode and sense transmitted raw sensor data packets, decode the raw sensor data, and display it on the OLED.

(1) To test the power regulator, batteries were attached and it was connected to an electronic load. Load current was swept from 10 mA to 500mA, and power efficiency was logged. The maximum expected current

draw occurs when the SX1262 transmits, with a maximum expected value of 118mA [3, Table 3-5]. The Handheld must also support current draw from the OLED display and MCU in parallel to the transmitter, so more than 118mA is expected. As such, testing to 500mA provides an overestimate to prove the functionality of our LM2576 regulator under operational load.

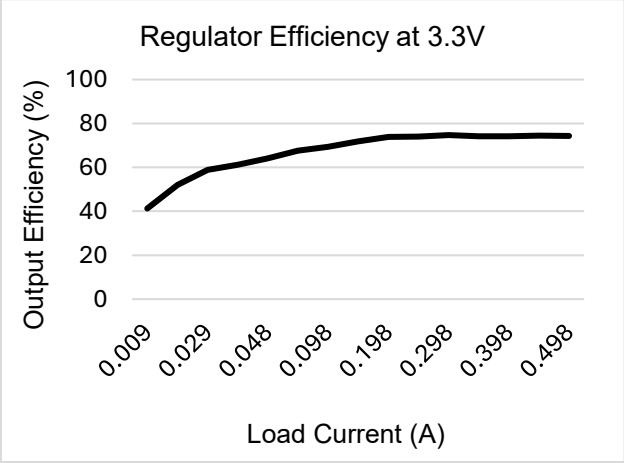


Fig. 8. Power efficiency of the LM2576 regulator swept from 17 to 500Ω. Maximum power efficiency is seen at 28Ω.

(2) Activation of the MCU and supporting components was tested by constructing a simple program to toggle an onboard GPIO pin using the external crystal oscillators for measurement by an oscilloscope, as in V: *Environmental Sensor* section (1). The handheld transceiver was connected to the regulator and turned on. Oscilloscope measurements of the GPIO pin confirmed activation of the MCU and frequency measurements confirmed stable oscillation of the external oscillators.

(3) Functionality of the OLED and, consequently, I2C, was tested by constructing a simple program writing a basic UI to the OLED screen. Power was applied and observation of the programmed text on the display confirmed operation of the OLED.

(4) Functionality of the SX1262 SDR and, consequently, SPI, was tested using the FHSS algorithm developed for the SX1262. The transmitter was configured to output its maximum power of +22dBm using the PA settings recommended by its manufacturer, Semtech. The SX1262 output port was connected to an FPC1000 spectrum analyzer with 50Ω termination to mimic an antenna. Max hold trace mode was used to sample the time-averaged peak power output across all hopping channels, the results of which are shown in Fig. 10. These measurements confirmed operation of the SX1262, albeit at a lower power level than configured. Multiple SX1262 chips were tested using the same software load, all of which produced similar output powers of 18dBm.

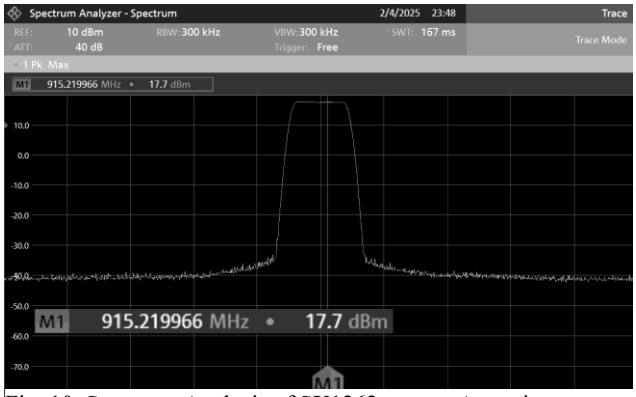


Fig. 10. Spectrum Analysis of SX1262 output. A maximum output power of approximately 18 dBm is measured.

(5) To test power radiation from the antenna, a Yagi-Uda antenna was connected to the output of the SX1262. A patch antenna was then connected to the input of the energy harvesting board, and a multimeter was attached across the harvesting capacitors. The transmitter was then activated, and multimeter readout confirmed the capture of radiated power by the energy harvester.

(6) To test Bluetooth scanning, a program was constructed that enters a Bluetooth scanning mode on start-up, captures transmitted Bluetooth packets, decodes the raw values of temperature and humidity, and writes scanned values to the OLED. The environmental sensor was constructed as in V: *Environmental Sensor* section (3) on a DC supply such that it output Bluetooth packets every one second. Observation of the OLED display confirmed capture of Bluetooth data and comparison to a reference thermometer and hygrometer confirmed accurate sensor values.

Upon completion of software development and all 6 testing steps, the handheld transceiver was labeled complete with full functionality, and no further testing was performed.

VI. RESULTS

A Yagi-Uda antenna was attached to the completed handheld transceiver, and it was mounted on a test fixture. The environmental sensor was attached to the energy harvester, and a patch antenna was connected. The combined sensor module was then mounted on a test fixture such that the antenna-to-antenna (A2A) distance between the sections was 1m. To measure power transmission efficiency, an oscilloscope was attached to the capacitor bank on the sensor module. The transmitter was activated, and a timer was started. A cycle charging time of approximately 45 seconds was measured. Assuming a

power transmission of 18dBm, this corresponds to a transmission of approximately 2.84J. The capacitor voltage threshold was set to 4.3 V, near the maximum allowed by the AEM30330. Assuming a 220uF capacitor, this voltage corresponds to 2.034mJ. Overall transmission efficiency is thus 716ppm. A maximum charging distance of approximately 1.5m A2A was measured, after which the incident power was too low to activate the AEM30330. Charging rate was highly dependent on accurate aiming of the antennas; both Yagi Uda and patch antennas are highly directional, and orientation greatly affects transmission efficiency. At 1m A2A, a Bluetooth packet transmission catch rate of approximately 80% was measured. Temperature measurements were taken outside, in standard room conditions, and in a freezer. Accurate measurements were observed in all conditions compared to a reference thermometer-hygrometer.

VII. CONCLUSION

This project demonstrates the ability to convert 915MHz RF power into usable DC power, removing the need to provide a battery or any other power source to initiate environmental sensing. At only 18dBm (63mW) of radiated power, a charging distance of over a meter is demonstrated, and accurate environmental measurements are taken from a device with component lifespans in the decades. This represents a practical replacement for long-term environmental sensors. Strong proof of concept for energy harvesting circuits that operate on ambient AM/FM radio signals or other forms of stray RF power is also demonstrated. Thousands of radio towers transmit many kilowatts of power, most of which is simply lost to the environment. Energy harvesting projects such as this one demonstrate a way to make use of this wasted power. Further development of this project would include optimization of the harvesting circuit via simulation of the RF path. Conversion of the lumped-element network to a transmission line equivalent would minimize loss and allow for increased charging distance. Utilization of a true +36dBi EIRP radiation setup (SDR + antenna) would allow for a significant increase in charging distance. Alternate projects could demonstrate ambient harvesting, biomedical sensing, or other long-life IoT applications.

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BIOGRAPHY



Ryan Witt is a senior Electrical Engineering major at UCF with a focus on MEMS devices & applications. He currently works as a member of the UCF Dynamic Microsystems Lab under Dr. Reza Abdolvand and will be pursuing his Ph. D in Electrical Engineering.



Jacob Henderson is an Electrical Engineering major at UCF with classwork focusing on analog and RF electronics. Currently an intern at Lockheed through UCF's CWEP program, he hopes to further his design and problem-solving skills through this project.



Nicholas Vakhordjian is an Electrical Engineering and Computer Science student at UCF, completing his BS/MS through the accelerated track. He works as an undergraduate teaching assistant and has interests in power electronics, power generation and software development.

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